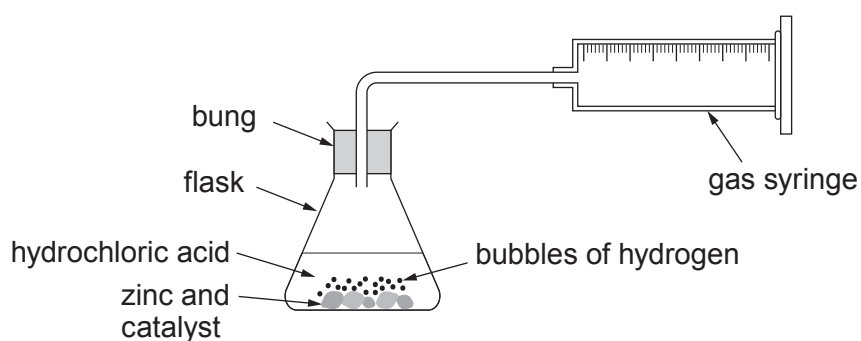


5. The diagram below shows apparatus that can be used to investigate the rate of the reaction between large zinc pieces and hydrochloric acid. A catalyst was also added to the flask.

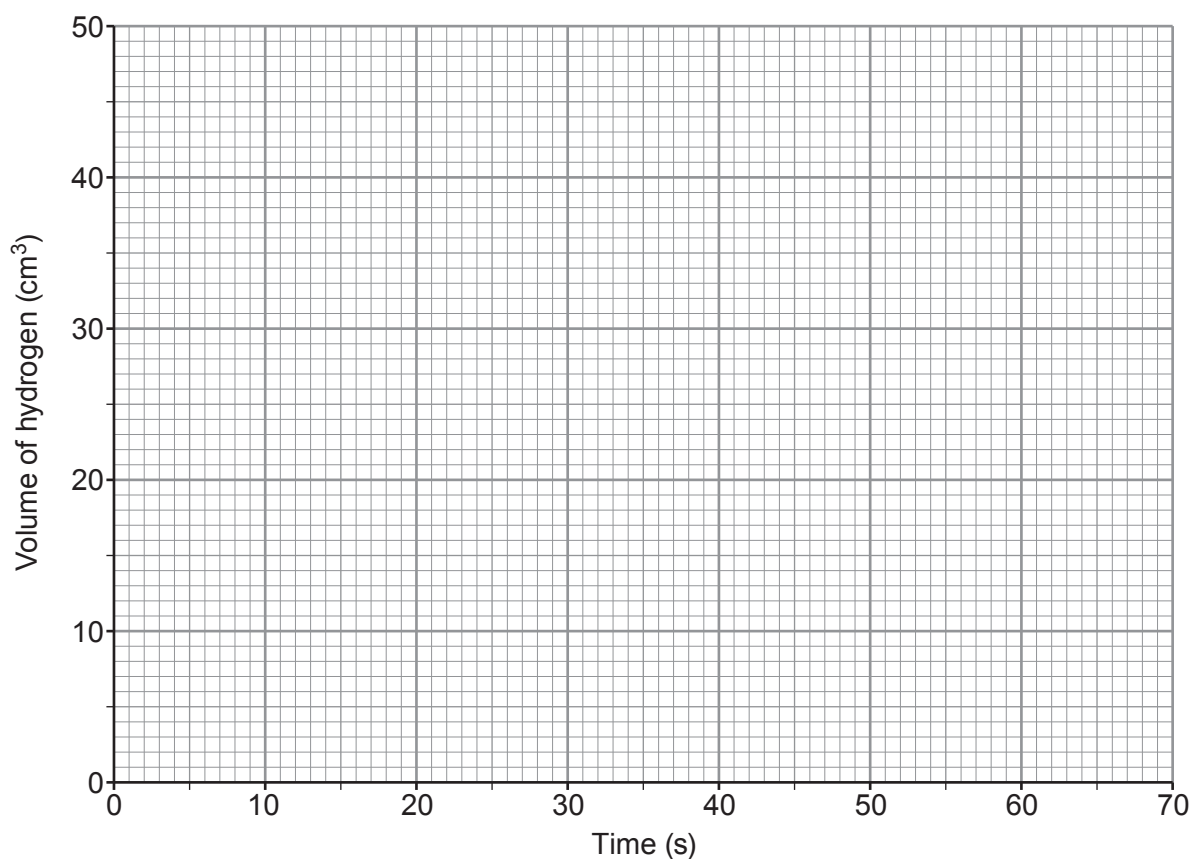


Large pieces of zinc were placed in excess dilute hydrochloric acid and the volume of hydrogen produced was recorded every 15 seconds. The results obtained are shown below.

Time (s)	0	15	30	45	60
Volume of hydrogen (cm ³)	0	28	40	48	48

- (a) (i) Plot the results on the grid below and draw a suitable curve.

[3]



(ii) Use your graph to answer the following questions.

I. Describe the relationship between volume of hydrogen produced and time.

[2]

.....

.....

.....

II. State the volume of hydrogen collected at 25 s.

[1]

volume = cm³

III. At what time did the reaction stop?

[1]

time = s

(b) (i) State the purpose of the catalyst.

[1]

.....

(ii) The mass of catalyst added at the start of the experiment was 1.5 g.
State the mass of catalyst remaining after 60 s.

[1]

mass = g

(iii) State how the **volumes in the table** would be different if the catalyst had not been added.

[1]

.....

(c) (i) State how the **volumes in the table** would be affected by using the same mass of zinc **powder** instead of large pieces of zinc.

[1]

.....

(ii) Explain how the **volumes in the table** would be affected if the original experiment was repeated at a higher temperature.

[2]

.....

.....

.....

